MockTIA LABS

NETWORK SUPPORT SERVICES / 4.2.11



MISSION: RESTORE A LINUX WORKSTATION TO THE CORPORATE NETWORK

Inspect the enp2s0 adapter, locate its persistent configuration, and correct the IPv4 details.

Apply the interface settings, verify address, gateway and DNS, then test reachability.

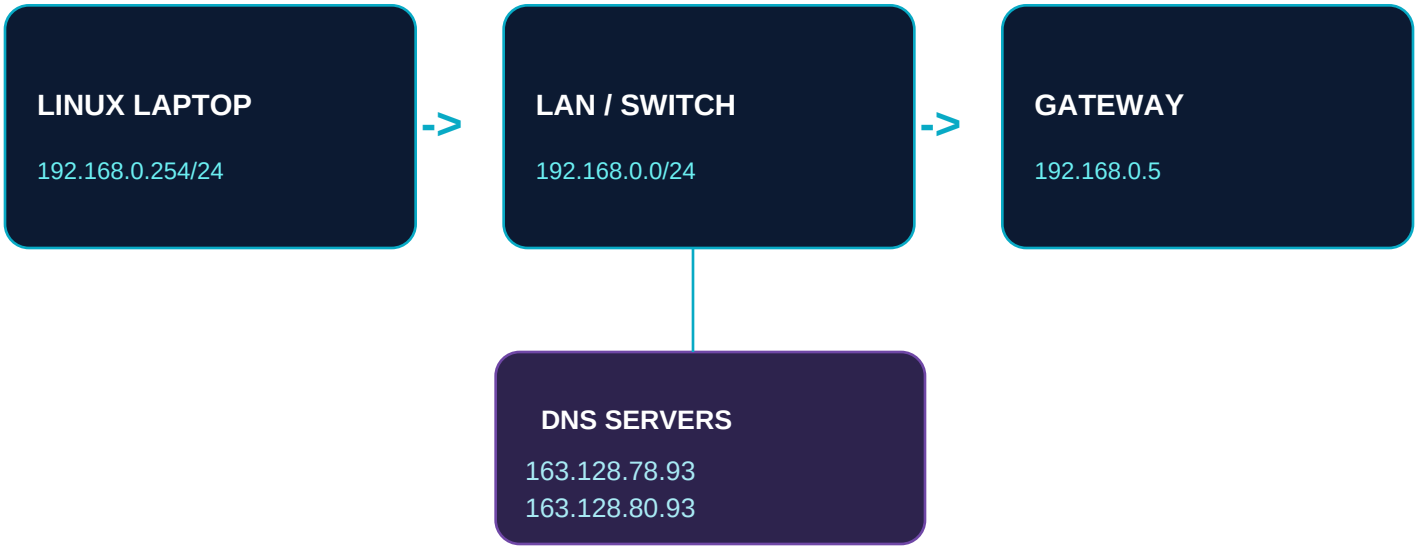
The graded simulation uses a legacy ifcfg file; modern Ubuntu commonly uses Netplan.

WORK ORDER REQUIREMENTS

Adapter / IPv4	enp2s0 / 192.168.0.254/24
Subnet / broadcast	255.255.255.0 / 192.168.0.255
Gateway	192.168.0.5
DNS (preferred / alternate)	163.128.78.93 / 163.128.80.93

Illustrated network reference

Know which devices and values belong in the Linux configuration



READ THE FIELDS

IP address	Uniquely identifies the laptop on the 192.168.0.0/24 LAN.
Subnet mask	255.255.255.0 means that the first 24 bits describe the subnet.
Broadcast address	192.168.0.255 targets all hosts on this local /24 broadcast domain.
Default gateway	192.168.0.5 forwards packets to destinations outside the LAN.
DNS resolver	Converts www.corpnet.xyz into an IP address; DNS is not routing.

Encapsulation reminder

Application data -> TCP/UDP segment -> IP packet -> Ethernet frame

Legacy Linux configuration walkthrough

Simulated /etc/sysconfig/network-scripts/ifcfg-enp2s0

1 DISCOVER THE INTERFACE

ip addr show

```
cd /etc/sysconfig/network-scripts && ls
```

2 EDIT THE PERSISTENT IFcfg CONFIGURATION

```
TYPE=Ethernet
BOOTPROTO=none
NAME=enp2s0
DEVICE=enp2s0
ONBOOT=yes
IPADDR=192.168.0.254
NETMASK=255.255.255.0
BROADCAST=192.168.0.255
GATEWAY=192.168.0.5
DNS1=163.128.78.93
DNS2=163.128.80.93
```

3 APPLY AND VERIFY

```
sudo systemctl restart NetworkManager

ip addr show | ip route | cat /etc/resolv.conf

ping -c 3 192.168.0.5

ping -c 3 163.128.78.93

ping -c 3 www.corpnet.xyz
```

Linux vocabulary

Reference definitions for troubleshooting and certification preparation

NIC	Network interface card; network adapter such as enp2s0.
IPv4	32-bit logical network address.
CIDR	Prefix notation, e.g., /24 for 255.255.255.0.
Subnet	Set of hosts with a shared network prefix.
Broadcast	Single frame or packet addressed to all hosts on a broadcast domain.
Gateway	Router used to reach remote IP subnets.
DNS	Name resolution service from names to IP addresses.
DHCP	Service that assigns network settings automatically.
BOOTPROTO	Legacy ifcfg property controlling how interface addressing is obtained.
ONBOOT	Legacy ifcfg property enabling the connection at startup.
ifcfg-enp2s0	Legacy configuration file for the enp2s0 network interface.
ip addr	Inspect IP interfaces and addressing.
ip route	Inspect the routing table and default gateway.
ping	ICMP Echo troubleshooting tool.
Netplan	Current Ubuntu networking YAML configuration tool.
NetworkManager	Service that manages network connections on many Linux systems.
resolv.conf	Resolver configuration file; can be generated and managed by systemd-resolved.

Review questions and troubleshooting

Complete the practical before submitting your assessment PDF

1. What prefix corresponds to 255.255.255.0?

2. Why must the interface gateway be on the same local subnet?

3. A ping to 192.168.0.5 works, but a hostname fails. What do you inspect?

4. What is the difference between IP changes made with ip addr and persistent settings?

5. Why might /etc/sysconfig/network-scripts not exist on a newer Ubuntu host?

TROUBLESHOOTING GUIDE

No gateway ping

Confirm correct IP/mask, link state and that 192.168.0.5 is reachable.

DNS IP works, website fails

Inspect nameserver fields and the resolver output.

Changes disappear after reboot

Recheck persistent config and the network service that manages the adapter.

Network access lost after applying

STUDENT CHECKLIST

[ ] Network settings verified [ ] Gateway and DNS tested [ ] Hostname resolved

[ ] Knowledge check complete [ ] Score checked [ ] Assessment saved as PDF

Modern Ubuntu reference

A modern comparison — not the scored legacy lab workflow

Netplan: /etc/netplan/*.yaml

```
network:
  version: 2
  renderer: networkd
  ethernets:
    enp2s0:
      dhcp4: false
      addresses: [192.168.0.254/24]
      routes:
        - to: default
          via: 192.168.0.5
      nameservers:
        addresses: [163.128.78.93, 163.128.80.93]
```

Modern validation commands

```
sudo netplan try  # apply temporarily with rollback prompt
sudo netplan apply  # persist changes after verification
ip addr show enp2s0
ip route show
resolvectl status
```

Ubuntu documentation: https://ubuntu.com/server/docs/configuring-networks/